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A broad learning model with dual-path feature encoding for robust and expedient incremental fault diagnosis

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ABSTRACT

Intelligent fault diagnosis (IFD) models that rely on one-shot learning often struggle in dynamic mechanical systems because they exhibit limited knowledge stability and high computational cost under incremental learning (IL). To address the trade-off between robustness and efficiency, this paper proposes a Multi-Scale High-Resolution and Source-Informed Broad Learning Model (MHRSI-BLM). By leveraging the computational efficiency of broad learning for rapid weight correction and pseudo-inverse updating, the proposed framework achieves efficient incremental updates. In addition, the MHRSI-BLM framework adopts a dual-path feature-encoding strategy to improve robustness: a multi-scale high-resolution pathway enhances feature representation, while a complementary source pathway provides a relatively stable feature anchor for incremental updates. Experiments on two rolling-bearing datasets demonstrate that the proposed method achieves strong diagnostic accuracy with competitive computational efficiency.

1. Introduction

Rolling bearings are critical components in modern industrial machinery and equipment, as maintaining their healthy operating state is essential for ensuring system stability (Liang et al., 2023). Operating continuously under relatively harsh environmental conditions, they are prone to various failure modes that can ultimately compromise system stability and lead to significant economic losses (Zhou et al., 2021). Timely monitoring and diagnosis of incipient bearing faults, followed by appropriate countermeasures, can prevent further damage to mechanical components (Chen et al., 2025b). Consequently, intelligent fault diagnosis (IFD), which aims to maintain the stable operation of rolling bearings, has attracted sustained attention from researchers worldwide.

Traditional fault diagnosis methods, such as Empirical Mode Decomposition (EMD) (Ke et al., 2022), Multivariate Variational Mode Decomposition (MVMD) (Song et al., 2022), Cluster-Weighted Envelope Spectrum (CWES) (Chen et al., 2025a) and Empirical Wavelet Transform (EWT) (Xu et al., 2020), mainly rely on complex signal processing techniques. Despite their progress, these approaches often require extensive parameter tuning and rely heavily on prior knowledge, which introduces uncertainty and may reduce diagnostic reliability. Consequently, data-driven methods like Deep Learning (DL), particularly Convolutional Neural Networks (CNN) (Chen et al., 2023; Song et al., 2024; Xu et al., 2024a; Zhu et al., 2025), have been widely adopted for their power-

ful automated feature-learning capabilities. However, in real-world scenarios, mechanical equipment frequently operates under varying conditions, leading to nonstationary data distributions. Most existing DL models assume a fixed data distribution based on static datasets, resulting in significant performance degradation when encountering distribution shifts. To address fault diagnosis under variable working conditions, numerous models have been proposed. Yu et al. (2023) proposed a novel fault diagnosis model based on a Graph Neural Network (GNN) combined with a Dynamic Graph Embedding (DGE) mechanism, enabling the model to adapt to fluctuating operating conditions with excellent domain adaptation capability. Jin et al. (2025) enhanced domain-specific adversarial neural networks by incorporating a multi-channel, multi-scale CNN-LSTM-ECA feature extraction module, achieving accurate fault diagnosis of rolling bearings under varying operating conditions. Feisa et al. (2025) proposed an enhanced one-dimensional deep CNN method that attained high accuracy in identifying multiple health states of rolling bearings across diverse operating conditions. Shahzad et al. (2025) introduced an Entropy-Enhanced Kronecker Convolutional Feature Pyramid (EKCFP) approach to further improve diagnostic accuracy under variable working conditions.

The aforementioned methods, relying on the batch learning paradigm, introduce fundamental operational constraints. First, batch learning's reliance on static historical datasets severely compromises the model's capacity to generalize to novel data patterns or shifting

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operational conditions. This inevitable lack of adaptability results in missed diagnoses, which can rapidly escalate into severe system failures and significant economic losses. Second, batch learning creates inherent operational discontinuity. Acquiring new data requires the immediate suspension of service and a full retraining process. The full retraining process induces a substantial monitoring gap, preventing continuous, real-time fault detection. Such a discontinuous approach is fundamentally unsuited for dynamic industrial environments. IL offers a viable alternative, enabling the retention of prior knowledge while continuously integrating new information. Liu et al. (2023) proposed a Lifelong Learning Method Based on Generative Feature Replay (LLMGFR) that combines real features from the current task with generated features to effectively address classifier imbalance and mitigate catastrophic forgetting, thus enabling continuous learning in dynamic environments. Wang et al. (2023) introduced an Incremental Learning-based Multi-Task Shared Classifier (IL-MTSC) for bearing fault diagnosis under diverse operating conditions, which is capable of preserving and expanding knowledge. Zhang et al. (2024) introduced a Deep Adaptive Sparse Residual Network (DASRN) employing a task-aware dynamic masking strategy to regulate the retention and utilization of weights, balancing memory capacity and learning ability while alleviating catastrophic forgetting.

DL has achieved remarkable success in static closed-set fault diagnosis tasks. However, its practical deployment in continuous incremental industrial scenarios is fundamentally constrained by two inherent limitations. On the one hand, mainstream DL models rely on computationally intensive, backpropagation-driven iterative optimization. This creates severe bottlenecks for real-time incremental model updates on resource-constrained industrial edge devices. On the other hand, the inherent architectural rigidity of static DL frameworks hinders the seamless integration of novel fault knowledge. Such frameworks typically either incur catastrophic forgetting of pre-learned fault patterns during lightweight incremental updates, or require full-model retraining with hundreds of iterative epochs to accommodate new fault knowledge, leading to extremely long update cycles, excessive computational overhead, and an inability to support low-latency continuous updates in industrial online diagnosis scenarios. To address these critical challenges, the Broad Learning System (BLS) (Chen & Liu, 2018) has emerged as a promising alternative framework. Building on this efficiency, several variants have been introduced for fault diagnosis, such as the Improved Lifelong Learning IFD Broad Learning System (IBLS) (Fu et al., 2023) and the Causal Broad Learning Model (CBLM) (Xu et al., 2024b), which significantly reduce time consumption while maintaining accuracy. However, traditional BLS relies on features generated by randomly initialized weights for its feature and enhancement nodes. This non-deterministic feature generation inherently fails to capture the crucial information required for effective bearing fault representation, diminishing its utility in accurately modeling fault evolution under complex operating conditions. This unguided, randomized feature extraction lacks the capacity to precisely encode subtle fault characteristics, resulting in poor separability between features of new and old fault patterns. This limitation renders the diagnostic model susceptible to collapse, compromising both accuracy and robustness, and ultimately failing to meet high-reliability industrial demands.

To address the robustness-efficiency dilemma in IL for complex rolling-bearing fault diagnosis, this paper proposes the Multi-Scale High-Resolution Source-Informed Broad Learning Model (MHRSI-BLM). The methodology processes time-frequency representations derived via Short-Time Fourier Transform (STFT). The Multi-Scale High-Resolution Squeeze-and-Excitation (MHRSE) network is strategically employed to mitigate spatial information decay and channel redundancy, thereby enhancing the robustness of feature representation. Computational efficiency is supported by the BLM's rapid weight correction and pseudo-inverse update mechanisms, enabling rapid model updates. Crucially, a complementary Source Layer is introduced, which functions as a stable feature anchor to impose an intrinsic constraint on the feature space,

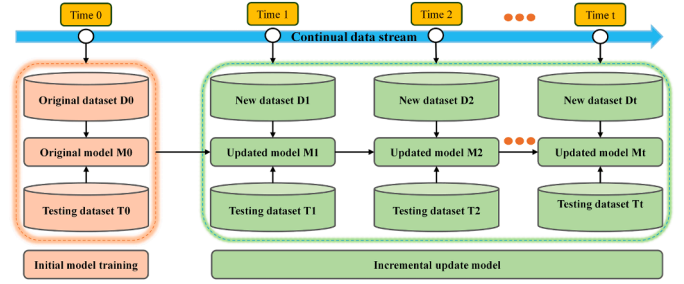


Fig. 1. Incremental learning process for different working conditions.

effectively mitigating catastrophic forgetting during IL and achieving robust dual-path feature encoding. This design enables MHRSI-BLM to provide a practical framework for continuous and efficient fault diagnosis under changing operating conditions. The main contributions of this paper are summarized below.

1) We propose the MHRSI-BLM model, an integrated knowledge-driven incremental learning architecture designed to address the trade-off between diagnostic knowledge stability and computational efficiency in industrial systems.

2) We design a Multi-Scale High-Resolution Squeeze-Excitation network that alleviates spatial-information degradation and channel redundancy, thereby improving the robustness of feature representation.

3) We introduce a Source Layer that serves as a stabilizing component in the feature space. Empirically, it helps maintain the numerical stability of the overall feature matrix during incremental updates and mitigates catastrophic forgetting.

4) We validate the proposed framework on two benchmark datasets and show that it achieves robust incremental diagnostic performance with competitive computational efficiency.

The remainder of this paper is organized as follows. Section 2 provides a brief description of the problem and broad learning. Section 3 describes the proposed MHRSI-BLM methodology in detail. Section 4 presents the experimental validation and performance analysis using two rolling bearing datasets. Finally, Section 5 concludes the paper.

2. Preliminaries

2.1. Problem formulation

This paper focuses on fault diagnosis under domain-incremental scenarios. Typically, the initial model contains a dataset of health and failure conditions under a given operating condition. Let $D = \{D_n\}_{n=0}^N$ denote data collected from multiple operating conditions, where N denotes the number of operating conditions. For the initial task T_0 , the dataset $D_0 = \{(x_j^0, y_j^0); j = 1, 2, \dots, m\}$ is used to train the initial diagnostic model M_0 , where m denotes the number of samples, $x_j^0 \in X$ denotes the input signal, and $y_j^0 \in Y$ denotes the corresponding label. Assume that the total number of diagnostic tasks is T , where T_0 is the non-incremental task and T_t ($t = 1, 2, \dots, T-1$) denotes the incremental tasks. In the incremental task T_t , the new data streams collected under different operating conditions will be input into the model M_t . Model M_t is updated using the dataset D_t based on the previous model M_{t-1} , where $D_t = \{(x_j^t, y_j^t); j = 1, 2, \dots, m\}$. For datasets D_t at different stages, the label classes are the same, but the sample distributions are different. This paper aims to learn from multiple domain distributions, ensuring that knowledge of old domains is not forgotten while learning new domains. Fig. 1 shows the flowchart of the incremental training process of the model in a dynamically evolving operating environment of industrial equipment.

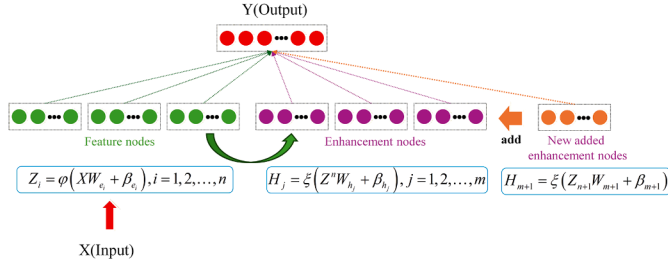


Fig. 2. Illustration of BLS.

2.2. Broad learning

The BLS comprises an input layer, a hidden layer, and an output layer, as illustrated in Fig. 2. Similar to other neural networks, the BLS is trained after the data are input and the model parameters are configured. Let X and Y be the input data and their corresponding labels, respectively. To extract the mapping features from the input data, X is transformed into the i -th set of mapping features, denoted by Z_i .

$$Z_i = \varphi(XW_{e_i} + \beta_{e_i}), i = 1, 2, \dots, n \quad (1)$$

where $\varphi(\cdot)$ represents the activation function, W_{e_i} and β_{e_i} represent the weight and bias of the i -th mapping of the feature node, respectively.

Let $Z^n = [Z_1, \dots, Z_n]$ denote the first n sets of mapped features. Similarly, the j -th set of enhancement features is denoted as

$$H_j = \xi(Z^n W_{h_j} + \beta_{h_j}), j = 1, 2, \dots, m \quad (2)$$

where $\xi(\cdot)$ denotes the activation function, and W_{h_j} and β_{h_j} denote the weight and bias of the j -th mapping of the enhancement node, respectively.

The first m sets of enhancement features are denoted by $H^m = [H_1, \dots, H_m]$. The output of the BLS can be represented as follows.

$$\begin{aligned} Y &= [Z_1, Z_2, \dots, Z_n | H_1, H_2, \dots, H_m] W^m \\ &= [Z^n | H^m] W^m \\ &= A^m W^m \end{aligned} \quad (3)$$

where $A^m = [Z^n | H^m]$ denotes the merged feature nodes and enhancement nodes. W^m denotes the output weight matrix, which can be calculated by Eq. (4).

$$W^m = (A^m)^+ Y \quad (4)$$

where $(A^m)^+$ denotes the pseudo-inverse of the feature matrix. According to ridge-regression theory, the output weights can be written as

$$W = (\lambda I + A A^T)^{-1} A^T Y \quad (5)$$

where λ and I denote the regularization parameter and the identity matrix.

3. Proposed method

3.1. Multi-scale high-resolution squeeze-and-excitation network

The MHRSE module aims to address the two core bottlenecks in deep feature extraction: spatial information degradation and channel dimension redundancy, thereby providing BLM with robust and separable feature representations. This module combines the high-resolution preservation structure $\mathcal{F}_{HR}$ with the explicit channel recalibration mechanism $\mathcal{G}_{SE}$. This module serves as the feature mapping layer of the BLM, responsible for providing highly robust, high-resolution features. The design of MHRSE is a structural optimization for the non-iterative feature mapping requirements of the MHRSE-BLM framework. It avoids the hierarchical structure of traditional CNN backbones designed for loss function optimization, and instead focuses on providing high-quality and stable feature tensors for pseudo-inverse computation.

Under complex operating conditions, vibration signals of rolling bearings are highly susceptible to strong noise. The Short-Time Fourier Transform (STFT) converts these vibration signals into two-dimensional spectrograms, which enables a structured depiction of fault-related information and provides a foundation for subsequent feature recognition. The STFT can be mathematically expressed as:

$$STFT\{x[n]\}(m, k) = \sum_{n=-\infty}^{\infty} x[n]w[n - mH]e^{-j\frac{2\pi kn}{M}} \quad (6)$$

where m is the time index, k is the frequency index, $w[n]$ is the window function, H is the step size, and M is the number of FFT points.

The STFT can convert fault signals into spectrograms, but the features it generates have limited capacity for characterizing weak faults, making them inadequate for direct fault diagnosis. CNNs are often used to solve this problem. However, when processing spectrograms, CNNs lose crucial spatial details due to multiple downsampling steps, consequently leading to the complete loss of subtle fault features in high-resolution feature maps. High-Resolution Net (HRNet) (Sun et al., 2019) addresses the information degradation issue of traditional CNNs through parallel branches, but its single-scale convolution at the input layer still presents limitations. To overcome this, we propose an improved HRNet that integrates multi-scale convolutions at the input layer to simultaneously capture both local details and global context. Let C_{base} be the base number of channels. The kernel size is set to $k \in \mathcal{K} = \{3, 5, 7\}$ with the number of convolution kernels fixed at 32, and the convolution operation $F_{k \times k}$ comprises the weight tensor W_k and bias b_k . Details on the selection of kernel size and quantity are provided in Section 4.3.2.

$$\mathcal{Y}_k = F_{k \times k}(X; W_k, b_k) \in \mathbb{R}^{C_{base} \times F \times T} \quad (7)$$

where F and T represent the spatial dimensions of the input feature map, corresponding to the frequency and time axes of a 2D time-frequency map.

Subsequently, the three parallel feature maps are concatenated along the channel dimension to obtain the fusion tensor $\text{Concat}(\mathcal{Y}_3, \mathcal{Y}_5, \mathcal{Y}_7) \in \mathbb{R}^{F \times T \times 3C_{base}}$.

MHRSE uses HRNet as its feature extraction backbone to overcome the spatial information loss problem in traditional downsampling networks. This structure contains N parallel branches $\mathcal{B}^l = \{\mathcal{B}_1^l, \mathcal{B}_2^l, \dots, \mathcal{B}_N^l\}$, where $\mathcal{B}_i^l$ denotes the feature tensor corresponding to the i -th branch at the l -th stage, satisfying the resolution relationship $F_1 \times T_1 > F_2 \times T_2 > \dots > F_N \times T_N$. The overall architecture is illustrated in Fig. 3.

At each stage l , semantic enhancement of multi-scale information is achieved through bidirectional feature fusion. The output $\mathcal{B}_i^{(l)}$ of the i -th branch is composed of the combined contributions of the aligned re-sampling $\mathcal{R}_{j \rightarrow i}$ of all branches and the 1×1 -channel aligned convolution W_{ji} .

$$\mathcal{B}_i^{(l)} = \sum_{j=1}^N W_{ji} * \mathcal{R}_{j \rightarrow i}(\mathcal{B}_j^{(l-1)}) \quad (8)$$

where $\mathcal{R}_{j \rightarrow i}(\cdot)$ is the resolution adaptation function. When $j < i$, $\mathcal{R}_{j \rightarrow i}$ is a strided convolution to achieve downsampling; when $j > i$, $\mathcal{R}_{j \rightarrow i}$ is a bilinear interpolation to achieve upsampling. After L stages of fusion, the feature maps from all branches are first aligned to the same spatial resolution and then fused to obtain the final high-resolution feature tensor U , which preserves both local detail and multi-scale contextual information.

$$U = \sum_{i=1}^N \mathcal{C}_i^{(L)} \quad (9)$$

During the initial construction stage, each branch generates feature channels, typically structured with N channels in the high-resolution branch and $2N, 4N$, and $8N$ in the parallel low-resolution branches. Under complex operating conditions, the contribution of these feature channels varies significantly, and signals are often contaminated with

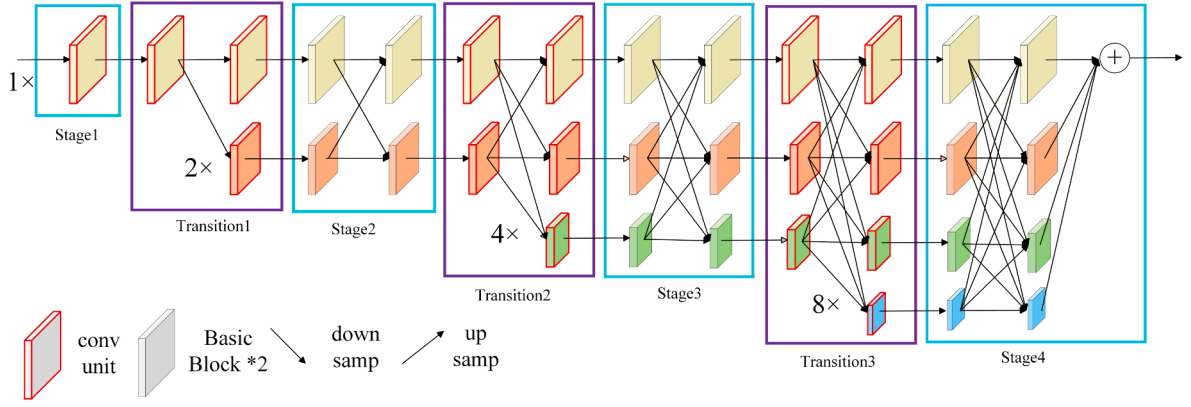


Fig. 3. Architecture of the proposed HRNet.

noise that degrades diagnostic accuracy (Wang et al., 2021). Traditional SENet (Hu et al., 2018) only performs recalibration after the output of a single convolutional layer, and its global context aggregation is limited by the local receptive field of that layer. MHRSE applies SENet to the fused tensor U generated by the F_{HR} mechanism, thereby using multi-scale global context to compute more accurate and robust channel weights.

For the fused feature tensor U , let $u_c \in \mathbb{R}^{F \times T}$ denote the feature map of the c -th channel. Global average pooling is then applied over the frequency and time dimensions to obtain the channel descriptor z_c .

$$z_c = \frac{1}{FT} \sum_{i=1}^F \sum_{j=1}^T u_c(i, j) \quad (10)$$

By stacking all channel descriptors, we obtain $z = [z_1, z_2, \dots, z_C]^T$. The channel attention vector s is then computed by a nonlinear gating function with two trainable weight matrices W_1 and W_2 .

$$s = \sigma(W_2 \delta(W_1 z)) \quad (11)$$

where W_1 and W_2 are trainable weight matrices, $\delta(\cdot)$ denotes the ReLU activation function, and $\sigma(\cdot)$ denotes the sigmoid activation function.

The final output F_c is obtained by scaling u_c through channel-wise multiplication using the channel activation coefficient s_c .

$$F_c = u_c \cdot s_c \quad (12)$$

Let Θ_{MHRSE} denote all trainable parameters in the MHRSE module, including the multi-scale convolution parameters W_k, b_k in Eq. (7), the branch-fusion and channel-alignment parameters W_{ji} in Eq. (8), the convolutional parameters in the basic blocks, and the squeeze-and-excitation parameters W_1, W_2 in Eq. (11). The temporary linear softmax classifier is parameterized by $\Sigma = \{W_c, b_c\}$. During construction of the initial model, Θ_{MHRSE} and Σ are jointly optimized once on the initial dataset D_0 by supervised back-propagation. Specifically, the globally pooled MHRSE feature $f_{\Theta_{MHRSE}}(X_i)$ is connected to the temporary classifier, and the prediction and joint optimization objective are formulated as

$$\hat{p}_i = \text{softmax}(W_c f_{\Theta_{MHRSE}}(X_i) + b_c) \quad (13)$$

$$\begin{aligned} \min_{\Theta_{MHRSE}, \Sigma} \mathcal{L}_{MHRSE} = & -\frac{1}{N_0} \sum_{i=1}^{N_0} \sum_{q=1}^C y_{i,q} \log \hat{p}_{i,q} \\ & + \lambda_{\Theta} \|\Theta_{MHRSE}\|_2^2 + \lambda_{\Sigma} \|\Sigma\|_2^2 \end{aligned} \quad (14)$$

where N_0 is the number of initial training samples, C is the number of health-state classes, $y_{i,q}$ is the one-hot label, and λ_{Θ} and λ_{Σ} are the weight-decay coefficients for the MHRSE module and the temporary classifier, respectively. The minimization in Eq. (14) explicitly indicates that Θ_{MHRSE} and Σ are optimized jointly during the initial

representation-learning stage. After convergence, the temporary classifier is removed and Θ_{MHRSE} is frozen. The fixed MHRSE module is then used to extract features for both the initial task and all subsequent incremental tasks. During incremental updating, only the broad-learning output weights are computed or updated through ridge regression and the recursive pseudo-inverse mechanism in Eqs. (17)–(20); therefore, no iterative back-propagation through MHRSE is required after the initial representation-learning stage. This two-stage strategy combines a trainable feature extractor with low-cost incremental updates.

In this way, informative channels are emphasized while less useful ones are suppressed, yielding more discriminative feature representations for subsequent broad-learning-based mapping.

By preserving high-resolution details through the F_{HR} module, the $\mathcal{G}_{SE}$ module uses a more informative basis for weight allocation. This complementary mechanism reduces the feature-information loss and channel-weight misalignment commonly observed in traditional single-branch convolutional feature extractors. This synergistic effect improves the model's ability to recognize fault features beyond that achieved by either module alone.

3.2. Incremental formulation of the broad learning model

For the initial dataset D_0 , the input is fed into the MHRSE feature extraction network to obtain the output feature f , and then the feature node Z_n and the enhancement node H_m are obtained through Eq. (1) and (2).

Although the MHRSE parameters are frozen after initial training, changes in operating conditions alter the activation statistics of the fixed feature extractor. Consequently, the derived embeddings Z_n and H_m may still undergo domain-induced distribution and orientation drift within the total feature matrix A during continuous incremental learning, which necessitates a dedicated stabilization mechanism. The Source Layer addresses this by serving as a structurally stabilizing kernel and a crucial low-level feature anchor. By extracting the untransformed feature vector S_l directly from the raw input, it effectively bypasses the semantic drift and homogenization inherent in shared convolution kernels. The integration of S_l into the final feature set A imposes a mathematically quantifiable intrinsic low-dimensional constraint on the deep feature space. This strategic constraint prevents the excessive entanglement of feature representations for distinct fault modes, thereby helping preserve the long-term structural integrity and stability of the diagnostic knowledge across all successive incremental stages.

The design rationale underpinning the prioritization of low-frequency steady-state information in the Source Layer is rooted in the inherent physical characteristics of rolling bearing vibration signatures. For rotating machinery, low-frequency components characterize the relatively steady energy distribution of the mechanical system, which exhibits stronger invariance across diverse health conditions. Conversely,

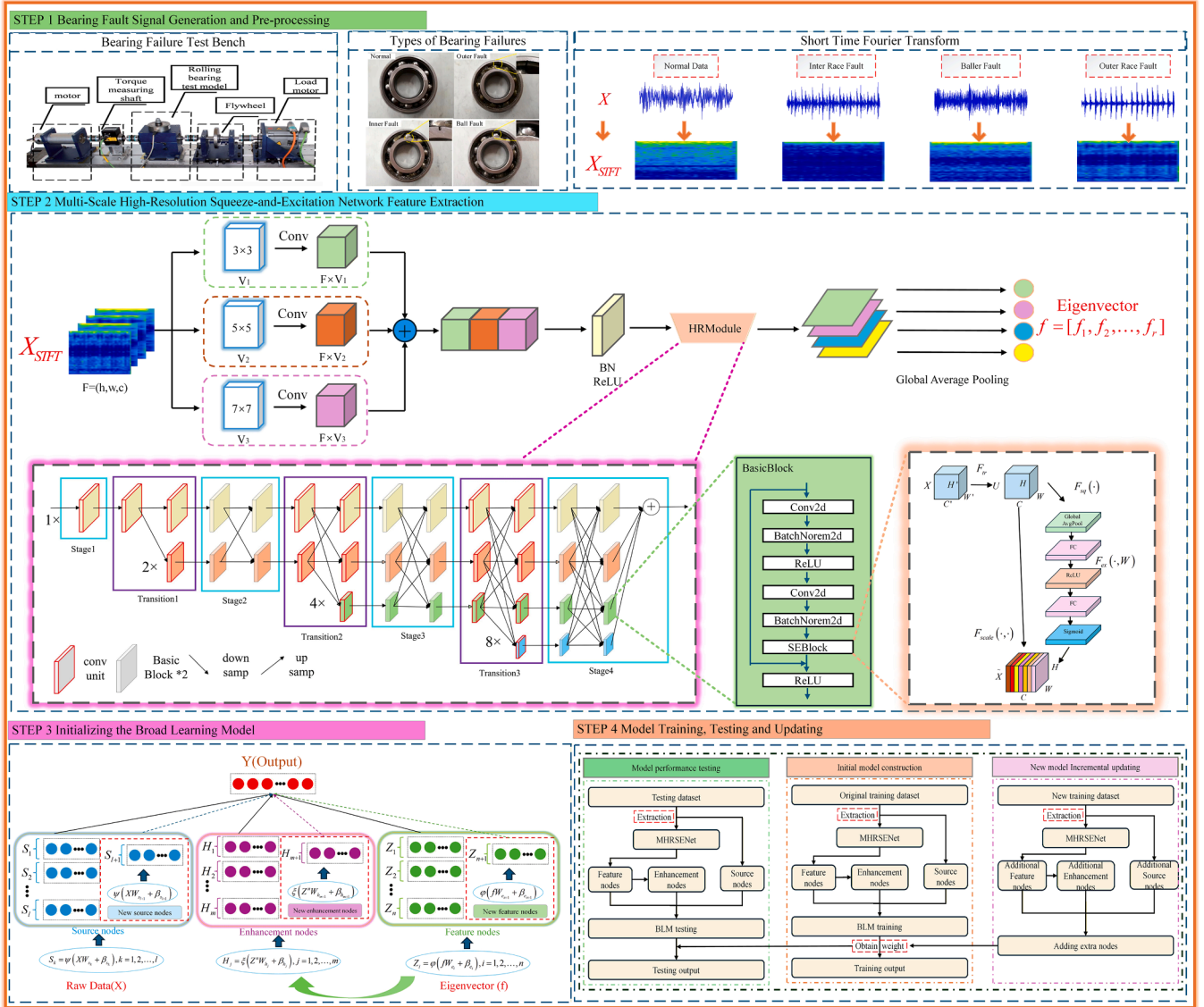


Fig. 4. Overall structure.

high-frequency bands contain critical transient impulses induced by localized bearing defects, yet are inherently non-stationary and highly vulnerable to ambient noise and transmission path modulations. In the incremental learning paradigm, over-reliance on such volatile high-frequency transients can induce overfitting to task-specific noise, which in turn exacerbates catastrophic forgetting. By anchoring the Source Layer to steady-state low-frequency spectral components, the model establishes a robust representation baseline that encodes the equipment's invariant dynamic characteristics. This preserves the stability of global domain knowledge, while enabling subsequent layers to adapt to discriminative high-frequency fault features introduced in successive incremental tasks (Liu et al., 2025).

$$S_k = \psi(XW_{s_k} + \beta_{s_k}), k = 1, 2, \dots, l \quad (15)$$

where $\psi(\cdot)$ denotes the activation function, W_{s_k} and β_{s_k} denote the weight and bias of the k -th mapping of the source node, respectively. Let $S^l = [S_1, \dots, S_l]$ denote the first l sets of source features.

In the framework proposed in this paper, the robustness of incremental learning is closely related to the numerical stability of the incremental weight W updates. This stability is affected by the ill-conditioned nature of the total feature matrix $A = [Z^n | H^m | S^l]$, which is mainly influenced by the homogenization of the deep features f .

Let A_k be the feature matrix of the k -th stage, and A_Δ be the newly added data block. Then the matrix is updated as follows:

$$A_{k+1} = \begin{bmatrix} A_k \\ A_\Delta \end{bmatrix} \quad (16)$$

The incremental mechanism of BLM relies on the efficient computation of new pseudo-inverses A_{k+1}^+ . This recursive formula is based on a series of key auxiliary matrices P, T, R :

$$P = I - A_k A_k^+ \quad (17)$$

$$T = A_\Delta A_k^+ \quad (18)$$

$$R = A_\Delta P \quad (19)$$

Eqs. (17)–(19) formulate the recursive update of the Moore-Penrose inverse. Eq. (17) defines the orthogonal projection matrix P to identify the null space of the existing features. Eq. (18) calculates the transformation matrix T , representing the knowledge redundancy between new and historical data. Eq. (19) derives the innovation matrix R , which extracts the "pure incremental information" from the new samples. The final recursive formula for A_{k+1}^+ is:

$$A_{k+1}^+ = \begin{bmatrix} A_k^+ - A_k^+ A_\Delta^T R^+ T \\ R^+ T \end{bmatrix} +$$

$$\begin{bmatrix} P(R^\top R)^\top (A_\Delta P P^\top A_\Delta^\top)^\top + A_\Delta P \\ R^\top (A_\Delta P P^\top A_\Delta^\top)^\top + A_\Delta P \end{bmatrix} A_{(k)}^+ \quad (20)$$

The numerical stability of Eq. (20) is strongly influenced by the minimum singular value $\sigma_{\min}(A_k)$ of A_k , while the condition number $\kappa(A_k)$ of A_k characterizes its degree of ill-conditioning.

$$\kappa(A_k) = \frac{\sigma_{\max}(A_k)}{\sigma_{\min}(A_k)} \quad (21)$$

When the deep features f become increasingly homogenized, $\sigma_{\min}(A_k)$ may approach zero, causing $\kappa(A_k)$ to increase sharply and making the pseudo-inverse update more sensitive to perturbations.

The Source Layer mechanism imposes an inherent regularization constraint on A_k by introducing a structurally invariant feature block. Since the extraction process of the matrix S^l is fixed, its minimum singular value is assumed to remain stable and nonzero throughout the incremental learning process.

$$\sigma_{\min}(S^l) \geq C_{\text{stable}} > 0 \quad (22)$$

S^l typically carries stable low-frequency information from the signal and is less sensitive to high-frequency noise and domain-induced variations in deep features. This physical characteristic provides the empirical basis for the stability assumption associated with C_{stable} .

The Source Layer introduces a relatively stable feature block into the overall feature matrix A_k . Because the source features are extracted through a fixed transformation, their numerical behavior is expected to be more stable than that of the deep features during incremental updates. In our framework, we therefore assume that the minimum singular value of the source feature block remains bounded away from zero.

Based on this observation, we use the following relation as a heuristic stability assumption:

$$\sigma_{\min}(A_k) \geq C_{\text{factor}} \cdot \sigma_{\min}(S^l) \geq C_{\text{stable}} \quad (23)$$

where C_{factor} is a projection factor that depends on the orthogonality of the feature space. Based on this observation, we use Eq. (23) as a heuristic stability assumption rather than as a general theorem for arbitrary block matrices. Mathematically, while the lower bound of the minimum singular value of an arbitrary block matrix is not universally constrained by its constituent sub-blocks, our empirical observations suggest that $\sigma_{\min}(A^k)$ maintains a stable proportional relationship with $\sigma_{\min}(S^l)$ in incremental learning scenarios. This phenomenon arises from the low redundancy between incremental data batches representing heterogeneous health states. As validated in existing research, maintaining a stable proportional relationship between the minimum singular values of

feature matrices is critical to mitigating model overfitting and ensuring numerical robustness across successive incremental learning stages (Wang et al., 2025c). Accordingly, adopting this proportionality condition enables a well-conditioned recursive update, further enhancing the model's robustness against numerical perturbations throughout the continuous incremental learning process.

By stabilizing $\sigma_{\min}(A_k)$, S^l constrains the condition number of $A_{(k)}$ to a controllable low range:

$$\kappa(A_{(k)}) \leq C_{\max} < \infty \quad (24)$$

According to the theory of relative pseudo-inverse perturbation, the upper bound of the relative error of the pseudo-inverse is proportional to the square of the condition number:

$$\frac{\|A_k^+ - (A_k + \Delta A)^+\|_F}{\|A_k^+\|_F} \leq \kappa(A_k)^2 \cdot \frac{\|\Delta A\|_F}{\|A_k\|_F} \quad (25)$$

where $\|\cdot\|_F$ denotes the Frobenius norm, and ΔA denotes the perturbation matrix.

According to pseudo-inverse perturbation theory, the relative error of the pseudo-inverse can be bounded by a term that depends on the condition number of A_k and the magnitude of the perturbation ΔA . Therefore, when the condition number of A_k is effectively controlled, the sensitivity of the pseudo-inverse update to perturbations can also be reduced. Consequently, the Source Layer helps limit error amplification during recursive updates and improves the numerical stability of the incremental learning process, rather than guaranteeing complete suppression of perturbation effects.

$$\frac{\|\Delta W\|_F}{\|W\|_F} \leq C_{\max}^2 \cdot \frac{\|\Delta A\|_F}{\|A_{(k)}\|_F} \quad (26)$$

Unlike residual connections (He et al., 2016) in conventional deep networks, the Source Layer in our framework is intended to improve the numerical stability of the recursive pseudo-inverse update by introducing a relatively stable feature component. This design helps mitigate catastrophic forgetting and improves the robustness of MHRSE-BLM in incremental learning.

3.3. Incremental learning framework based on MHRSE-BLM

The detailed algorithm process is shown in Table 1. The overall process of the proposed MHRSE-BLM algorithm, illustrated in Fig. 4, consists of four main steps:

Step 1: Bearing Fault Signal Generation and Pre-processing: Vibration signals are acquired from a bearing fault test bench. The raw signals

Table 1
Overall Incremental Learning Procedure of MHRSE-BLM.

Input: $D = \{D_n\}_{n=0}^{N-1}$, a data stream containing N sequential incremental stages.

Output: W : Output weight matrix; A^+ : Feature matrix pseudo-inverse.

Initialization: Jointly optimize Θ_{MHRSE} and Σ on D_0 using Eqs. (13) and (14); remove Σ and freeze Θ_{MHRSE} ; $A \leftarrow \emptyset$, $Y_{\text{total}} \leftarrow \emptyset$, $A^+ \leftarrow \emptyset$

1. **For** $n = 0$ **to** $N - 1$ **do**:
2. Calculate the spectrum X_Δ using Eq. (6)
3. Calculate deep features f_Δ using the frozen MHRSE module.
4. Calculate Z^Δ , H^Δ using Eqs. (1) and (2)
5. Calculate S^Δ using Eq. (15)
6. $A_\Delta \leftarrow [Z^\Delta \mid H^\Delta \mid S^\Delta]$
7. **If** $n = 0$ **then**
8. $A \leftarrow A_\Delta$; $Y_{\text{total}} \leftarrow y_n^l$
9. Calculate the pseudo-inverse matrix $A^+ \leftarrow \text{pinv}(A)$
10. **Else**
11. Calculate P , T , and R using Eqs. (17), (18), and (19).
12. $R^+ \leftarrow \text{pinv}(R)$
13. $A_{\text{old}}^+ \leftarrow A^+$
14. Update the pseudo-inverse A^+ according to Eq. (20).
15. $A \leftarrow [A^\top \mid A_\Delta^\top]^\top$
16. $Y_{\text{total}} \leftarrow [Y_{\text{total}}^\top \mid y_n^l]^\top$
17. **End If**
18. $W \leftarrow A^+ Y_{\text{total}}$
19. **End For**

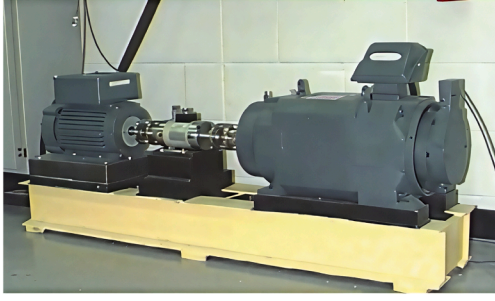


Fig. 5. Experimental data-acquisition platform at CWRU.

are preprocessed, and the STFT is applied to convert the vibration signals into spectrograms.

Step 2: Multi-Scale High-Resolution Squeeze-and-Excitation Network Feature Extraction: On the initial dataset D_0 , the MHRSE module and temporary classifier are jointly optimized using Eqs. (13) and (14). After this initial representation-learning stage, the classifier is removed and the frozen MHRSE module extracts feature vectors for all incremental stages.

Step 3: Initializing the Broad Learning Model: Initial weight matrices are computed using ridge regression by constructing feature nodes, enhancement nodes, and source nodes. Model capacity can be further enhanced by adding feature nodes and enhancement nodes.

Step 4: Model Training, Testing and Updating: By employing weight modification and pseudo-inverse matrix update mechanisms, the model parameters are rapidly updated from the initial model.

4. Experiments and results

The proposed method is validated on two rolling bearing datasets. The model is implemented in Python. Our experimental environment is configured with an Intel Core i7-11700 processor (8 cores, 16 threads, 2.5 GHz base frequency, 4.9 GHz maximum turbo frequency) and an NVIDIA RTX 3090 GPU. For the proposed framework and the MHRSE-based ablation variants, the reported stage-wise runtimes cover feature extraction and broad-learning training or updating at each stage, while excluding the one-time MHRSE pretraining cost.

4.1. Case 1: Bearing fault dataset from case western reserve university

4.1.1. Dataset description and division

To evaluate the proposed MHRSE-BLM method, we use rolling bearing vibration signals from the Case Western Reserve University (CWRU) Bearing Data Center. The experimental setup, shown in Fig. 5, uses an SKF-6205 drive-end bearing. Inner ring (IF), outer ring (OF), and ball (BF) failures were simulated with four distinct fault diameters (0.007 in to 0.028 in). Signals were sampled at 12 kHz. Table 2 summarizes the dataset, which comprises 10 bearing states under four different load conditions. Each load condition provides 2,400 samples, with an 8 : 2 split for training and testing. Each CWRU sample contains 1,024 consecutive time-domain points before STFT processing, which is one third of the 3,072-point sample length used in Case 2. The 10 states (labeled C0–C9) are uniformly distributed across subsets.

4.1.2. Comparison methods

Eleven methods, along with ablation studies, were selected for comprehensive performance evaluation, as detailed in Table 3. To provide a reasonably controlled comparison, the compared methods were implemented under the same data split and evaluation protocol. S1 serves as the baseline full-network fine-tuning, S2 (Loke et al., 2017) implements classical regularization-based IL, S3 (Li & Hoiem, 2017) employs knowledge distillation, and S4 (Lopez-Paz & Ranzato, 2017) uses a combined approach of exemplar memory and gradient constraints. For models S5

to S8, we replaced the MHRSE network with BaseNet, which is a simple two-layer convolutional structure. Furthermore, specialized IL methods were included: S9 (Wang et al., 2025b) utilizes a trainable domain predictor, S10 (Wang et al., 2025a) focuses on non-sample replay, and S11 (Jeeveswaran et al., 2024) employs a gradual adaptation strategy. All comparative models were adaptively implemented based on their core architectures and incremental principles, ensuring a consistent comparison on the rolling-bearing dataset used in this study.

A1 uses the basic BLM model with BaseNet and serves as the common baseline. A2 retains the complete A1 configuration, including the BLM, feature nodes, enhancement nodes, and BaseNet, and additionally introduces the Source component. It is therefore referred to as the Source-enhanced baseline rather than a model containing only the Source Layer. A3 removes the Source Layer and replaces BaseNet with the proposed MHRSE network, thereby evaluating the MHRSE enhancement without the Source component. MHRSE-BLM combines MHRSE with the Source Layer. This construction ensures that the A1-versus-A2 comparison evaluates the incremental contribution of the Source component while keeping the remaining baseline architecture unchanged, whereas the A1-versus-A3 comparison evaluates the contribution of MHRSE under the same incremental-learning protocol.

To further evaluate the computational efficiency of various incremental learning schemes, a comprehensive computational complexity analysis was performed. Computational overhead is an important metric for assessing the practicality of incremental learning algorithms, particularly in scenarios involving streaming data or resource-constrained edge computing. To ensure an objective comparison, we first unified the mathematical notations for model architecture parameters, data scales, and algorithm-specific hyperparameters. Table 4 provides a detailed definition of the mathematical symbols used in this analysis.

4.1.3. Experimental results and analysis

The window sizes for the BLM's feature, enhancement, and source nodes were set to 20, 10, and 5, respectively. The number of nodes per window was consistently maintained at 20. These experimental parameters remained consistent throughout the entire process. The source node layer extracts a 100-dimensional feature vector S^l . Concurrently, high-resolution features f are mapped to generate a 400-dimensional feature node Z^n and a 200-dimensional enhancement node H^m . These are concatenated into a 700-dimensional feature set $A_m = [S^l | Z^n | H^m]$, which is used to train the initial model M_0 . For incremental model updates, the models are efficiently derived by computing the pseudo-inverse of the new feature matrix A'_m and adjusting the weight matrix accordingly. As shown in Table 5, MHRSE-BLM exhibits high diagnostic accuracy: the initial model M_0 achieves 100.00%, while the newly constructed model at the final stage, M_3 , achieves 99.46%. Fig. 6 further validates the stability of the proposed method. The initial model achieved an outstanding 100% diagnostic accuracy, and subsequent incrementally updated models maintained a comparably high level of performance. Error analysis revealed that diagnostic misclassifications were predominantly localized to fault categories 4, 5, and 8. Furthermore, faults associated with larger defect diameters showed a higher propensity for misclassification. Notably, the updated models not only achieve exceptional performance on new samples but also maintain diagnostic accuracy exceeding 98.93% for historical samples. This successful prior knowledge preservation capability ensures stability and generalizability across different operational conditions.

The comparative experimental results are presented in Figs. 7 and 8 and Table 6. The non-IL baseline S1 shows substantial accuracy degradation and severe catastrophic forgetting. Regularization and distillation methods S2 and S3 confirm the effectiveness of mitigating forgetting, but their accuracy degrades significantly in later stages, particularly as S2 fails to address higher-order terms. S4 surpasses S1–S3, demonstrating notable forgetting mitigation through its gradient space constraint mechanism, yet still exhibits observable decay. All S1–S4 required 50 training iterations, leading to substantial computational costs. Models

Table 2
Details of the CWRU dataset.

Fault diameter (in)	0				0.007			0.021			0.028			Load
	NC	IF	OF	BF	IF	OF	BF	IF	OF	BF	IF	OF	BF	
Labels	0	1	2	3	4	5	6	7	8	9				
Data1	240	240	240	240	240	240	240	240	240	240	240	240	240	0 hp
Data2	240	240	240	240	240	240	240	240	240	240	240	240	240	1 hp
Data3	240	240	240	240	240	240	240	240	240	240	240	240	240	2 hp
Data4	240	240	240	240	240	240	240	240	240	240	240	240	240	3 hp

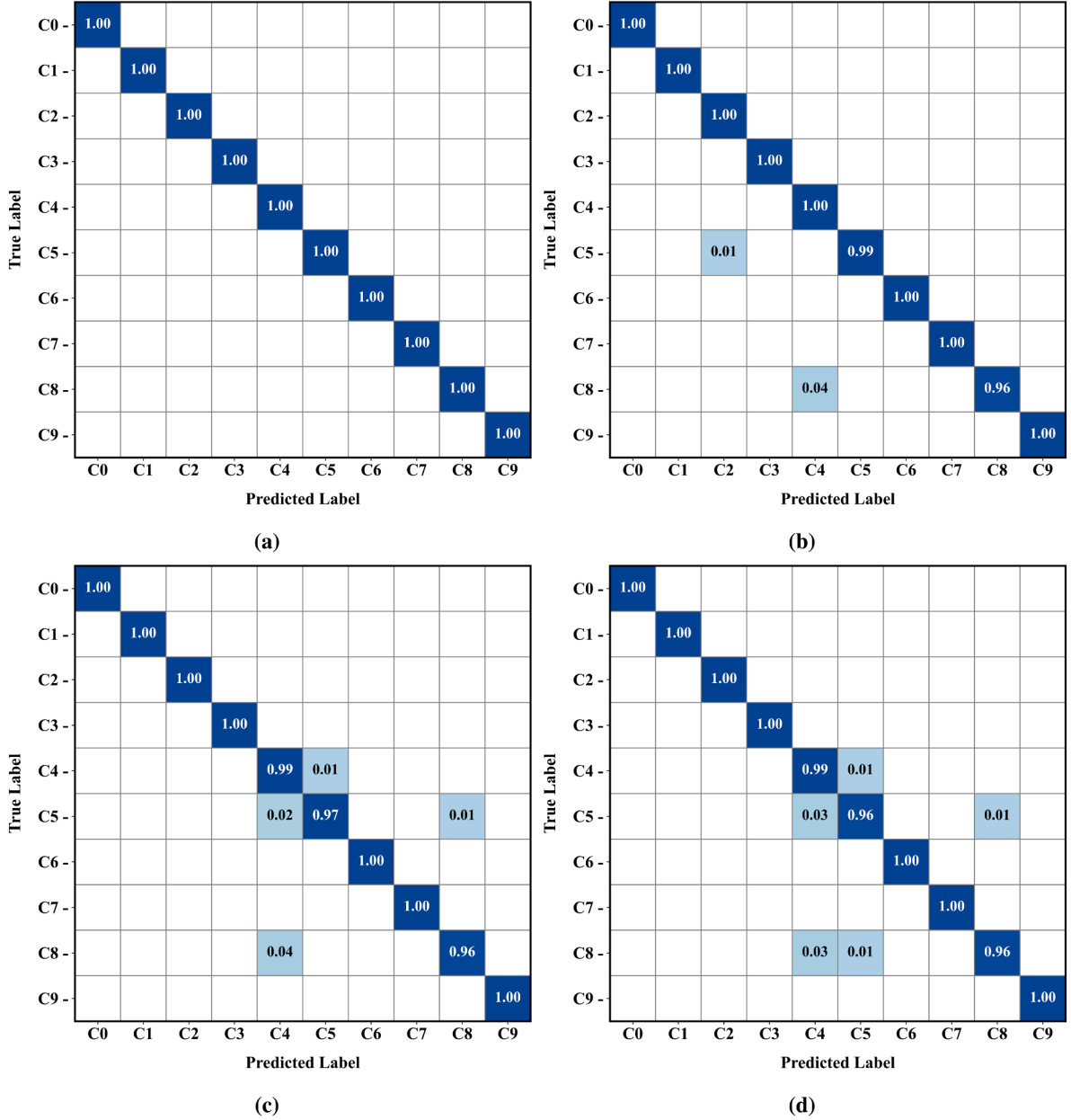


Fig. 6. Confusion matrix for the CWRU dataset. (a) 0HP. (b) 1HP. (c) 2HP. (d) 3HP.

S5-S8 replaced the MHRSE architecture with simpler CNNs to reduce computational cost. Although increasing iterations to 200 improved convergence, the simplification resulted in a slight accuracy decrease, despite achieving a significant reduction in runtime. S9 relies heavily on external domain detectors, showing the most significant degradation over long incremental sequences, dropping sharply to 95.17% by the fourth stage. S10 exhibits smaller decay but is limited by its univer-

sal feature alignment, preventing optimal accuracy. S11 uses gradual adaptation, but strict replay buffer constraints critically hinder its ability to retain complex fault knowledge, leading to an accuracy decline. The proposed MHRSE-BLM model achieves optimal accuracy across all stages. Crucially, it maintains computational efficiency comparable to CNN-based models, achieving the lowest runtime among all compared methods at Time 2.

Table 3
Compared methods.

Method	Description
S1	Fine-tuning
S2	EWC
S3	LWF
S4	GEM
S5	Fine-tuning-BaseNet
S6	EWC-BaseNet
S7	LWF-BaseNet
S8	GEM-BaseNet
S9	SOYO
S10	PINA
S11	DARE
A1	BLM + BaseNet
A2	A1 + Source
A3	BLM + MHRSE (without Source)
MHRSI-BLM	The proposed method

Table 4

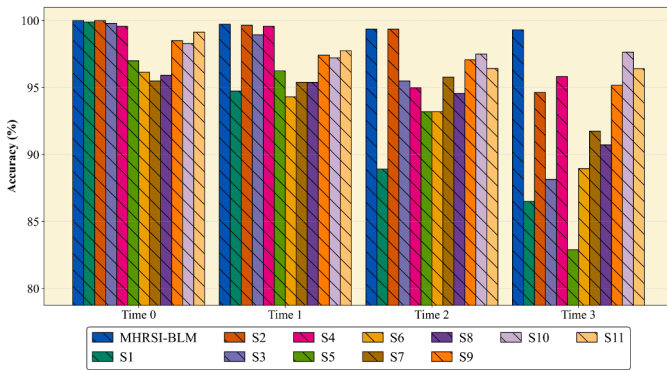
Definitions of symbols.

Symbol	Definition
N	Total number of training samples at the current stage
E	Number of training iterations
B	Batch size
t	Current stage index
P	Total number of parameters in the base model
$O(M)$	Computational complexity of one forward and backward pass per sample
L	Total number of neurons in BLS

Table 5

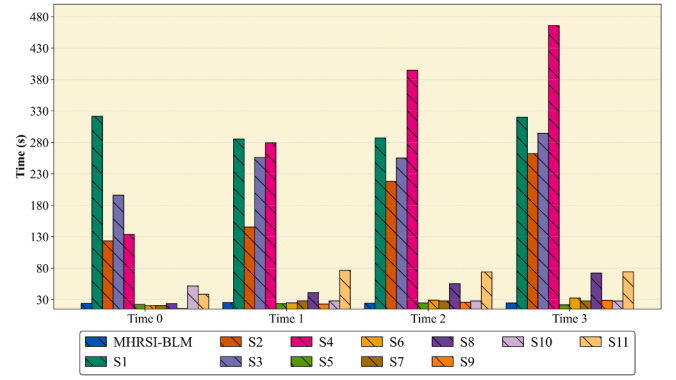
Testing accuracy of initial and updated models.

	Time 0	Time 1	Time 2	Time 3
	Accuracy	Accuracy	Accuracy	Accuracy
M0	100.00%	99.79%	99.57%	99.36%
M1	–	99.64%	98.93%	98.93%
M2	–	–	99.57%	100.00%
M3	–	–	–	99.46%

**Fig. 7.** Diagnostic accuracy of compared methods and MHRSI-BLM on CWRU.

As shown in Table 6, the accuracies of S4 and S10 do not decrease monotonically over time. In particular, their accuracies at Time 3 are slightly higher than those at Time 2. This phenomenon may be related to positive backward transfer, or more generally to feature-space realignment during later-stage learning. One possible explanation is that the features introduced by later tasks are more compatible with previously learned patterns, which may partially refine the global representation and improve earlier decision boundaries. As a result, later-stage learning may partially offset earlier interference and lead to a local recovery in diagnostic accuracy for previously learned tasks.

The computational advantage of MHRSI-BLM mainly originates from its broad-learning-based recursive update strategy rather than from

**Fig. 8.** Diagnostic runtime of compared methods and MHRSI-BLM on CWRU.

the deep feature extractor alone. In conventional deep incremental learning methods, the dominant training cost is typically of the order $O(E \cdot N \cdot M)$. For regularization- or constraint-based methods such as EWC and GEM, the complexity is further increased to $O(E \cdot N \cdot M + E \cdot \frac{N}{B} \cdot t \cdot P)$ due to the additional computation of Fisher-based penalties, gradient constraints, or stage-dependent historical information. By contrast, MHRSI-BLM reformulates the incremental update as a recursive pseudo-inverse computation in the broad learning space, with a dominant complexity of $O(N \cdot M + N \cdot L^2 + L^3)$. This formulation removes the epoch-dependent factor E and avoids the accumulation of stage-dependent constraint terms, thereby replacing repeated optimization of high-dimensional network parameters with one-pass feature extraction and low-dimensional matrix operations. Moreover, the recursive update in Eqs. (17)–(20) only requires the newly arrived feature block, rather than retraining on the entire historical dataset, which explains the nearly constant runtime across successive incremental stages. In addition, the Source Layer improves the conditioning of the global feature matrix by stabilizing its minimum singular value, as discussed in Eqs. (21)–(26), thereby enhancing the numerical reliability of the low-cost recursive update. Therefore, the proposed MHRSI-BLM achieves a more favorable trade-off between computational complexity, numerical stability, and diagnostic performance than the compared deep incremental learning methods.

t-SNE was further used to verify the classification performance of MHRSI-BLM by converting the extracted features into a two-dimensional feature representation, as shown in Fig. 9. Following training, feature scatter points of the same type exhibit tighter clustering than those of the original vibration samples. Features of different states are effectively separated, enabling accurate identification of fault types.

4.1.4. Ablation experiments

The ablation experiments aimed to quantitatively analyze the contributions of the MHRSE module to feature robustness and the Source Layer to the stability of IL knowledge by systematically deconstructing the key innovative modules of MHRSI-BLM. Three variants were set up, and accuracy and computational time were evaluated over four consecutive incremental phases from Time 0 to Time 3. The results are shown in Figs. 10, 11, and Table 7.

A2 retains the complete A1 configuration and additionally introduces the Source component; it does not retain only the Source Layer. A2 obtains accuracies of 83.42%, 87.11%, 87.76%, and 88.39% from Time 0 to Time 3, with an average accuracy of 86.67%. This is 5.42 percentage points higher than the 81.26% average accuracy of A1, suggesting that adding the Source component can improve the baseline with a relatively weak feature extractor by supplying additional stable low-level information. However, A2 remains substantially below A3 and the complete model. This result indicates that the Source Layer is intended as an auxiliary stability anchor and does not replace discriminative feature extraction. A configuration retaining only S^l while removing both the

Table 6

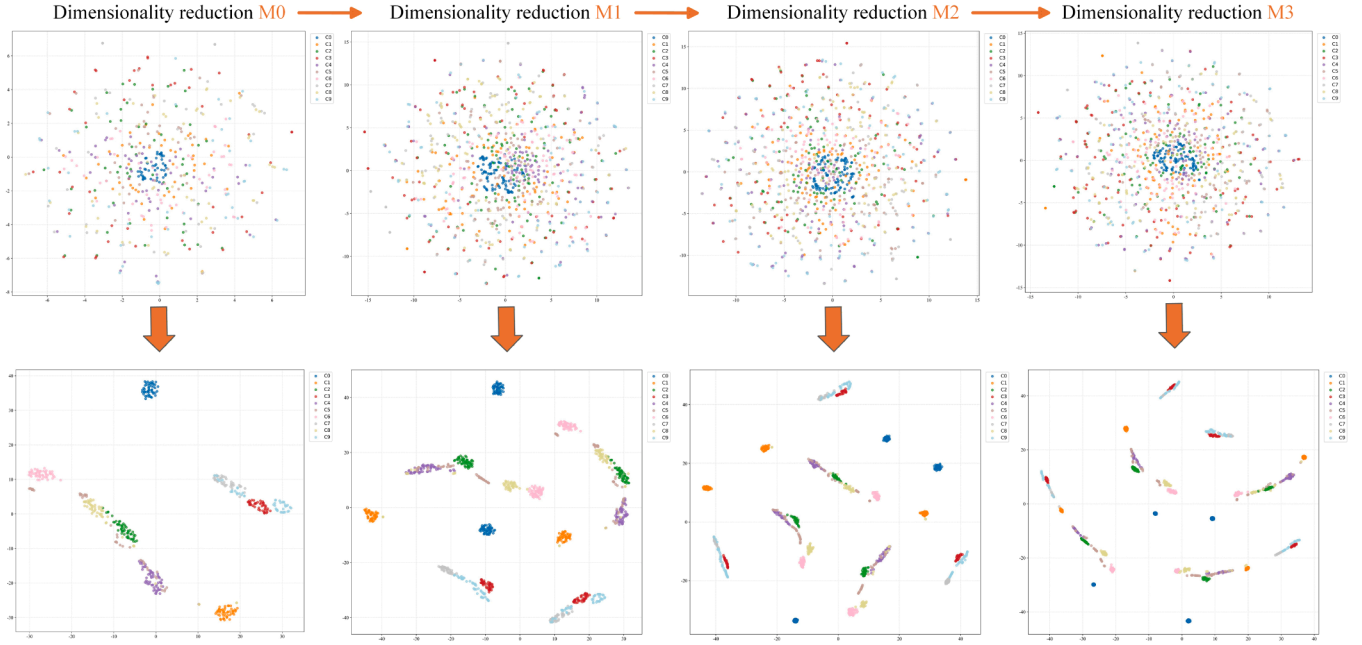
Accuracy and runtime of MHRSI-BLM and related methods.

	Time 0		Time 1		Time 2		Time 3		Time Complexity
	Accuracy	Time	Accuracy	Time	Accuracy	Time	Accuracy	Time	
S1	99.89%	321.54s	94.74%	285.62s	88.91%	287.07s	86.50%	320.30s	$O(E \cdot N \cdot M)$
S2	100.00%	123.50s	99.66%	145.54s	99.36%	218.19s	94.64%	262.35s	$O(E \cdot N \cdot M + E \cdot \frac{N}{B} \cdot t \cdot P)$
S3	99.79%	196.00s	98.93%	256.08s	95.49%	255.21s	88.14%	294.74s	$O(E \cdot N \cdot M)$
S4	99.57%	133.66s	99.57%	279.55s	94.99%	394.82s	95.82%	465.99s	$O(E \cdot N \cdot t \cdot M + E \cdot \frac{N}{B} \cdot t \cdot P)$
S5	97.00%	22.50s	96.24%	23.74s	93.20%	24.62s	82.89%	21.82s	$O(E \cdot N \cdot M)$
S6	96.14%	20.50s	94.31%	24.87s	93.20%	29.35s	88.95%	32.63s	$O(E \cdot N \cdot M + E \cdot \frac{N}{B} \cdot t \cdot P)$
S7	95.49%	20.74s	95.39%	28.21s	95.78%	27.94s	91.74%	27.82s	$O(E \cdot N \cdot M)$
S8	95.92%	23.68s	95.39%	41.07s	94.56%	55.35s	90.72%	72.34s	$O(E \cdot N \cdot t \cdot M + E \cdot \frac{N}{B} \cdot t \cdot P)$
S9	98.50%	15.95s	97.42%	24.13s	97.07%	25.91s	95.17%	29.00s	$O(E \cdot N \cdot M + E \cdot \frac{N}{B} \cdot t \cdot P)$
S10	98.28%	51.78s	97.21%	27.91s	97.50%	27.84s	97.64%	27.73s	$O(E \cdot N \cdot (M + M_{adp}))$
S11	99.14%	38.53s	97.75%	76.5s	96.42%	74.23s	96.41%	74.41s	$O(E \cdot N \cdot M)$
MHRSI-BLM	100.00%	24.20s	99.72%	25.50s	99.36%	24.34s	99.30%	24.52s	$O(N \cdot M + N \cdot L^2 + L^3)$

Table 7

Accuracy and runtime of MHRSI-BLM and ablation experiments.

	Time 0		Time 1		Time 2		Time 3	
	Accuracy	Time	Accuracy	Time	Accuracy	Time	Accuracy	Time
A1	78.14%	1.66s	82.90%	1.72s	82.54%	1.80s	81.44%	1.89s
A2	83.42%	2.85s	87.11%	2.54s	87.76%	3.13s	88.39%	3.01s
A3	95.21%	22.58s	93.39%	25.01s	92.64%	24.89s	90.73%	23.70s
MHRSI-BLM	100.00%	24.20s	99.72%	25.50s	99.36%	24.34s	99.30%	24.52s

**Fig. 9.** t-SNE visualization of the diagnostic features.

feature-node and enhancement-node paths would no longer preserve the intended BLM architecture. Accordingly, the A1-versus-A2 comparison is used to evaluate the incremental contribution of the Source component while keeping the remaining baseline architecture unchanged.

A3 uses MHRSE without the Source Layer and achieves 95.21%, 93.39%, 92.64%, and 90.73% across the four stages, corresponding to an average accuracy of 92.99%. Its Time-0 accuracy is 17.07 percentage points higher than that of A1, indicating that MHRSE substantially improves feature discrimination under the present setting. Nevertheless, its accuracy decreases by 4.48 percentage points from Time 0 to Time 3, suggesting that discriminative deep features alone may remain susceptible to domain-induced representation drift during incremental learning.

When both components are combined, MHRSI-BLM achieves 100.00%, 99.72%, 99.36%, and 99.30%, with an average accuracy of 99.60%

and a Time-0-to-Time-3 decrease of 0.70 percentage points. Under the current experimental protocol, these results suggest a complementary division of roles: MHRSE mainly contributes to feature discriminability, whereas the Source Layer helps improve incremental knowledge stability. Their combination achieves the strongest overall performance among the tested variants and provides a favorable robustness-efficiency trade-off in these experiments.

4.2. Case 2: Bearing fault dataset from paderborn university

4.2.1. Dataset description and division

To further evaluate the proposed method, we utilize the rolling-bearing dataset from the Paderborn University (PU) Bearing Data Center in Germany. The experimental test bench, shown in Fig. 12, comprises

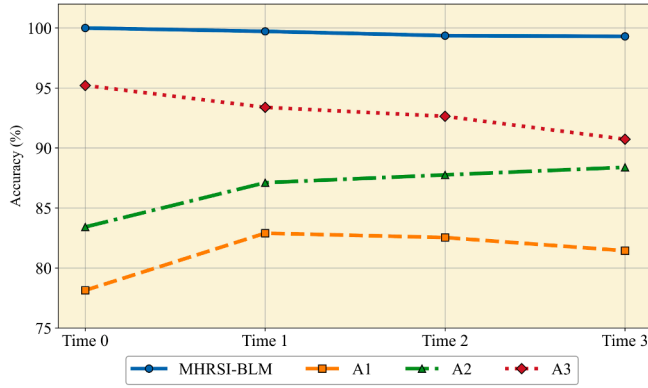


Fig. 10. Diagnostic accuracy of the ablation methods and MHRSI-BLM on CWRU.

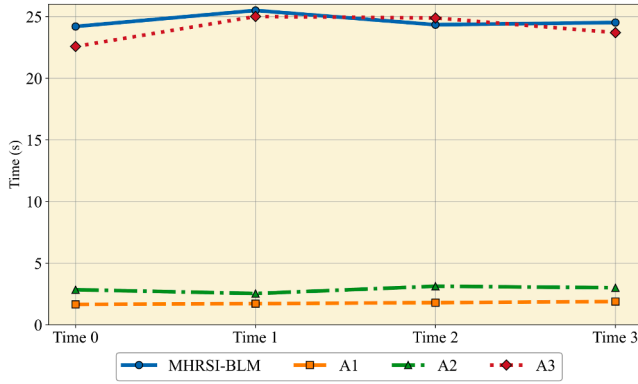


Fig. 11. Diagnostic runtime of the ablation methods and MHRSI-BLM on CWRU.

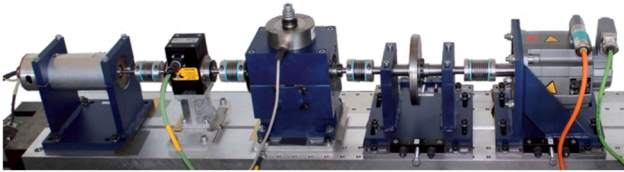


Fig. 12. Experimental data-acquisition platform at PU.

an electric motor, a torque shaft, and a rolling bearing test module. Faulty bearings were generated via both artificially induced damage and naturally evolved damage. Key parameters were monitored with varying sampling frequencies: 64 kHz for current and vibration signals, 4 kHz for mechanical parameters, and 1 Hz for temperature. The dataset covers four fault types: normal, outer ring failure (OF), inner ring failure (IF), and mixed outer and inner ring failure (MF). With a constant operating speed, three operating conditions were established based on varying load torques and radial forces, with the experimental temperature maintained between 45 and 50 °C. For processing, 12 types of samples were selected as detailed in Table 8. Each operating condition comprises 4,800 samples, with 3,840 samples allocated for training and 960 for testing. Each sample contains 3,072 data points.

4.2.2. Results summary

Because each sample contains more sampling points, the number of windows in each layer is set to 40, 20, and 10, while the number of nodes in each window remains 20. The diagnostic performance of the proposed MHRSI-BLM is presented in Table 9. The diagnostic accuracies of the initial model M0 and the updated models M1 and M2 are 99.89%, 99.39%, and 99.58%, respectively. Notably, the updated models demonstrate robust capability in distinguishing both new and historical samples, with

Table 8

Bearing codes and related information.

Bearing Code	Class Label	State Type	Damaged Part
K001	0	Normal	–
K002	1	Normal	–
KA01	2	Artificially induced damage	Outer
KI01	3	Artificially induced damage	Inner
KA03	4	Artificially induced damage	Outer
KA04	5	Naturally evolved damage	Outer
KI04	6	Naturally evolved damage	Inner
KI07	7	Artificially induced damage	Inner
KA16	8	Naturally evolved damage	Outer
KI18	9	Naturally evolved damage	Inner
KB23	10	Naturally evolved damage	Outer + inner
KB27	11	Naturally evolved damage	Outer + inner

Table 9

Testing accuracy of initial and updated models.

	Time 0	Time 1	Time 2
	Accuracy	Accuracy	Accuracy
M0	99.89%	98.48%	97.37%
M1	–	99.39%	98.58%
M2	–	–	99.58%

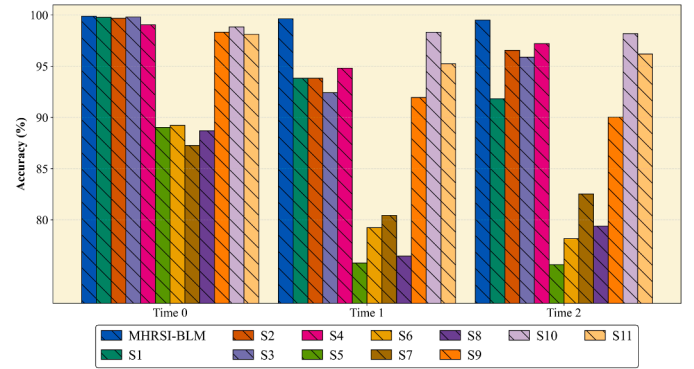


Fig. 13. Diagnostic accuracy of compared methods and MHRSI-BLM on PU.

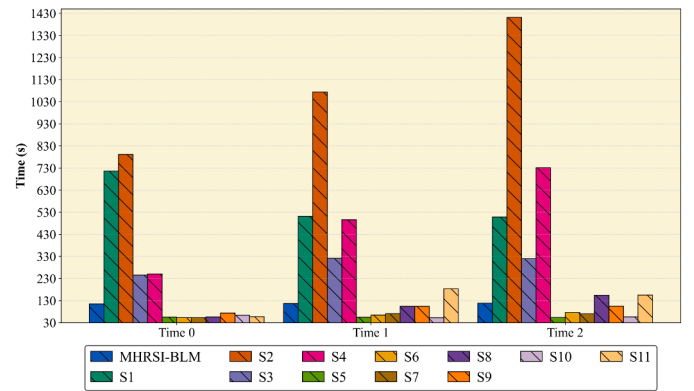


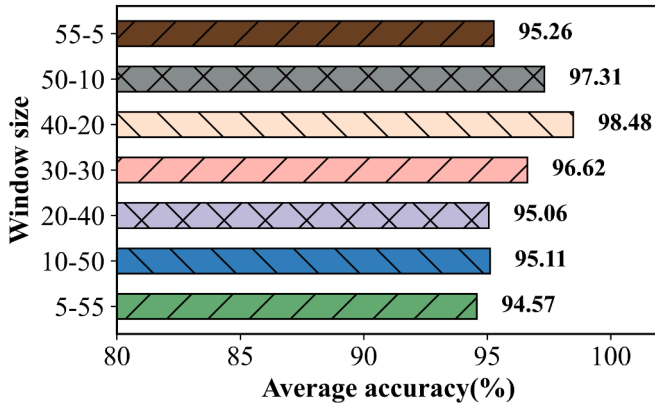
Fig. 14. Diagnostic runtime of compared methods and MHRSI-BLM on PU.

the accuracy for historical samples consistently exceeding 98.48%. Compared to the Case 1 dataset, the observed slight decrease in accuracy is attributed to the increased data complexity of the Paderborn University dataset, which contains three times as many sampling points per sample and an expansion from ten to twelve fault categories. Notwithstanding this greater complexity, the proposed model successfully maintains a robust diagnostic accuracy exceeding 99% under the current

Table 10

Accuracy and runtime of MHRSI-BLM and related methods.

	Time 0		Time 1		Time 2	
	Accuracy	Time	Accuracy	Time	Accuracy	Time
S1	99.79%	716.15s	93.83%	511.83s	91.83%	508.23s
S2	99.69%	791.11s	93.83%	1073.80s	96.55%	1412.11s
S3	99.81%	245.74s	92.42%	321.33s	95.89%	319.96s
S4	99.06%	250.39s	94.82%	495.74s	97.21%	731.06s
S5	89.02%	55.17s	75.78%	54.53s	75.63%	53.81s
S6	89.23%	53.15s	79.24%	64.62s	78.17%	75.99s
S7	87.26%	53.34s	80.44%	71.10s	82.53%	70.84s
S8	88.70%	55.97s	76.46%	103.93s	79.39%	154.47s
S9	98.33%	73.78s	91.95%	104.08s	90.03%	104.52s
S10	98.85%	63.48s	97.32%	52.24s	97.18%	56.53s
S11	98.12%	57.33s	95.24%	154.17s	96.20%	154.95s
MHRSI-BLM	99.89%	114.88s	98.93%	116.74s	98.51%	118.22s

**Fig. 15.** Average accuracy with different feature-node and enhancement-node window settings.

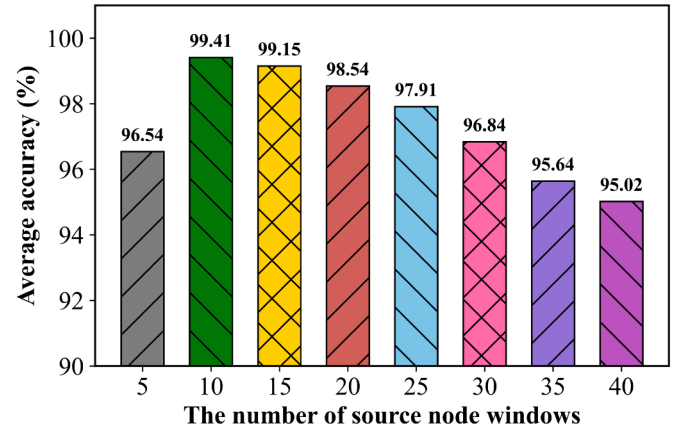
operating conditions, and retains an accuracy of over 97.37% for previously learned domains.

Similar to Case 1, the proposed MHRSI-BLM was compared with representative incremental learning methods under the same experimental setting, as shown in Table 10, Figs. 13 and 14. Overall, MHRSI-BLM achieves the best diagnostic accuracy among the compared methods on the PU dataset. At Time 2, its accuracy reaches 98.51%, which is higher than those of all baseline methods. In addition, the accuracies of the updated models remain stable, indicating that the proposed framework preserves previously learned knowledge more effectively than the competing approaches. Although the runtime of MHRSI-BLM is higher than that of some lightweight baselines, it remains substantially lower than that of several deep incremental-learning methods using the same backbone, while providing clearly better diagnostic performance. These results suggest that MHRSI-BLM achieves a favorable balance between accuracy, stability, and computational efficiency on the PU dataset.

4.3. Hyperparameter analysis

4.3.1. Exploration of window parameters

The diagnostic accuracy of the proposed model is influenced by the configuration of the BLM nodes. Using average accuracy as the evaluation metric, targeted experiments were conducted to investigate the effect of different node settings on model performance. The number of source nodes was held constant, and the number of window nodes was consistently set to 20. The total initial capacity for feature nodes and enhancement nodes was fixed at 1,200. Seven configurations were then evaluated by varying the number of windows for the feature nodes and enhancement nodes. The tested window size ratios were: 5-55, 10-50, 20-40, 30-30, 40-20, 50-10, and 55-5. As illustrated in Fig. 15, the model exhibits optimal performance when the ratio of feature node

**Fig. 16.** Average accuracy with different source-node window settings.

windows to enhancement node windows is set to 2 : 1 (40-20). This superior outcome suggests that this specific configuration provides the best balance. The feature nodes possess sufficient capacity to stably and robustly represent the input data. The enhancement nodes effectively introduce necessary higher-order feature interactions without inducing overfitting, thereby maintaining model generalization.

To further explore the impact of the Source Layer capacity on overall model performance, targeted experiments were conducted where the feature node and enhancement node window sizes were fixed at 40 and 20, respectively. The initial number of source node windows was varied from 5 to 40 with a step size of 5. As shown in Fig. 16, optimal performance was achieved when the number of source node windows was set to 10. Deviations in either direction resulted in performance degradation. Insufficient source nodes provided inadequate original feature information, hindering the effective coverage of key dimensions by the feature- and enhancement-node mappings. Conversely, an excessive number of source nodes led to the expansion of the feature matrix A_m , increasing computational complexity and introducing the risk of overfitting. Moreover, excessive capacity reduced the effective contribution of the nonlinear augmentation nodes, consequently leading to performance decline.

4.3.2. Exploring convolution kernel size and number

To verify the optimality of the proposed multi-scale convolution configuration ($3 \times 3, 5 \times 5, 7 \times 7$) in the MHRSE module, a comparative study was conducted across five distinct kernel size combinations. All methods were tuned under the same data split and evaluation protocol to provide a consistent comparison. The diagnostic accuracies on the CWRU dataset are summarized in Fig. 17. As illustrated in Fig. 17, the proposed ($3, 5, 7$) multi-scale kernel configuration achieves a superior diagnostic accuracy of 99.43%, notably outperforming the single-scale baselines of ($3, 3, 3$) and ($7, 7, 7$) by 1.95 and 1.79 percentage points, respectively. This result validates the synergistic effect of multi-scale fusion in capturing complementary transient impulses and structural modulations. The performance advantage over the even-sized ($2, 4, 6$) group further underscores the importance of the center-alignment property inherent in odd kernels for precise feature localization, while the accuracy degradation observed in the ($3, 7, 11$) group to 97.80% suggests that excessively large receptive fields introduce redundant low-frequency noise and feature blurring, thereby confirming the ($3, 5, 7$) combination as the optimal balance for characterizing rolling bearing vibration signatures.

The number of convolution kernels also affects model performance. The average accuracy and runtime for different numbers of convolutional kernels are shown in Fig. 18. The experimental results show that the diagnostic accuracy peaks at 99.64% when the number of convolution kernels is set to 32. Beyond this point, further increasing the number of kernels to 64 and 128 leads to slight performance

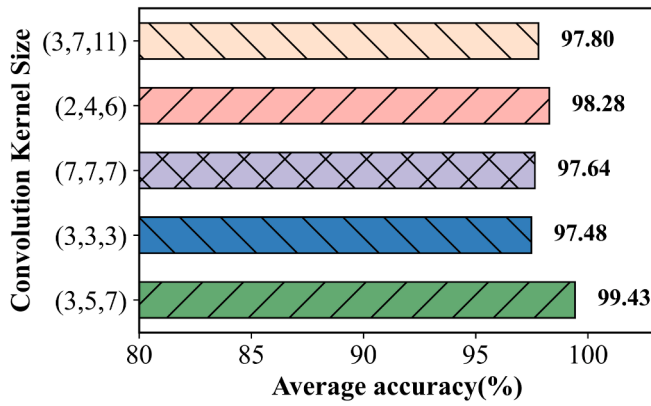


Fig. 17. Average accuracy for different kernel-size configurations.

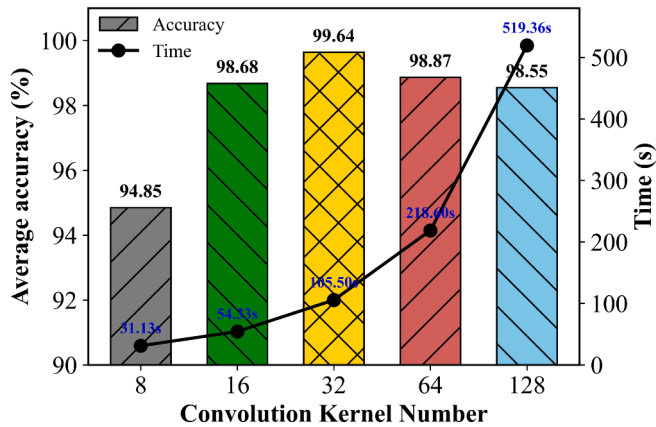


Fig. 18. Average accuracy and runtime with different numbers of convolutional kernels.

declines to 98.87% and 98.55%, respectively, suggesting that excessive kernels introduce feature redundancy and potential overfitting. Crucially, while the accuracy stabilizes after 32 kernels, the computational runtime exhibits a sharp, nonlinear increase from 105.50 s to 519.36 s, nearly quintupling the processing latency for a quadrupled kernel count. This significant escalation in temporal cost highlights the inherent trade-off between model complexity and operational efficiency, thereby identifying 32 kernels as an effective trade-off point that maximizes representative capacity for fault signature extraction while preserving the high-speed, real-time performance essential for incremental industrial diagnostics.

5. Conclusion

This paper proposes MHRSI-BLM, a broad-learning-based framework for incremental fault diagnosis under changing operating conditions. Experimental results on the CWRU and PU datasets show that the proposed method achieves consistently high diagnostic accuracy during incremental updates while maintaining competitive computational efficiency. The improvement mainly comes from two components: the multi-scale high-resolution feature extraction module, which enhances feature representation, and the Source Layer, which helps stabilize the incremental update process. Nevertheless, the interpretation of the Source Layer as a singular-value stabilizer remains largely empirical and should not be viewed as a fully general theoretical conclusion. Future work will focus on strengthening the theoretical analysis and validating the proposed framework in more complex, large-scale, and multi-modal industrial scenarios.

Despite its promising performance, the proposed method still has several limitations that warrant further investigation. First, the multi-branch feature extraction architecture contributes to a relatively high computational complexity. Second, the model relies on horizontal network expansion for knowledge retention, which may encounter memory constraints during very large IL data streams. To further improve scalability, future work will explore how the broad learning framework can be combined with more advanced incremental-learning strategies, such as replay-based or adaptation-based techniques. Furthermore, the current framework is primarily tailored for vibration signals. Recent studies have demonstrated the effectiveness of acoustic analysis and linear predictive coding in detecting abnormal conditions in induction motors (Głowacz et al., 2026). Inspired by this, future research will focus on generalizing the proposed architecture to multi-modal sensor fusion and extending its application to broader domains.

CRedit authorship contribution statement

Shengjie Zhang: Writing – original draft, Software, Methodology, Conceptualization; **Baoyi Xu:** Writing – review & editing, Validation; **Zeyun Yang:** Writing – review & editing, Supervision, Project administration; **Fen Wang:** Resources, Investigation; **Yuan Liu:** Formal analysis; **Zheng Xiang:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this study are publicly available.

References

- Chen, C. L. P., & Liu, Z. (2018). Broad learning system: An effective and efficient incremental learning system without the need for deep architecture. *IEEE Transactions on Neural Networks and Learning Systems*, 29(1), 10–24. <https://doi.org/10.1109/TNNLS.2017.2716952>
- Chen, S., Zheng, W., Xiao, H., Han, P., & Luo, K. (2023). A residual convolution transfer framework based on slow feature for cross-domain machinery fault diagnosis. *Neurocomputing*, 546, 126322. <https://doi.org/10.1016/j.neucom.2023.126322>
- Chen, T., Guo, L., Gao, H., Feng, T., & Yu, Y. (2025a). Clustering weighted envelope spectrum for rolling bearing fault diagnosis. *IEEE Transactions on Automation Science and Engineering*, 22, 3922–3932. <https://doi.org/10.1109/TASE.2024.3403665>
- Chen, Y., Yue, J., Liu, Z., & Chen, J. (2025b). A semi-supervised wise-attention weighted prototype network for rolling bearing fault diagnosis under noisy and limited labeled data conditions. *Neurocomputing*, 647, 130563. <https://doi.org/10.1016/j.neucom.2025.130563>
- Feisa, T. T., Gebremedhen, H. S., Kibrete, F., & Woldemichael, D. E. (2025). One-dimensional deep convolutional neural network-based intelligent fault diagnosis method for bearings under unbalanced health and high-class health states. *Structural Control Health Monitoring*, 2025, 3922–3932.
- Fu, Y., Cao, H., Chen, X., & Ding, J. (2023). Improved broad learning system for machinery intelligent fault diagnosis with increasing fault samples, fault modes, and running conditions. *ISA Transactions*, 136, 400–416. <https://doi.org/10.1016/j.isatra.2022.10.014>
- Głowacz, A., Czechowski, M., Brumerick, F., Guttén, M., & Głowacz, W. (2026). Abnormal noise detection of three-phase induction motor using linear predictive coding and neural networks. *Archives of Electrical Engineering*. <https://api.semanticscholar.org/CorpusID:284708016>
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *2016 IEEE Conference on computer vision and pattern recognition (CVPR)* (pp. 770–778). <https://doi.org/10.1109/CVPR.2016.90>
- Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. In *2018 IEEE/CVF Conference on computer vision and pattern recognition* (pp. 7132–7141). <https://doi.org/10.1109/CVPR.2018.00745>

- Jeeveswaran, K., Arani, E., & Zonooz, B. (2024). Gradual divergence for seamless adaptation: A novel domain incremental learning method. In *Proceedings of the 41st international conference on machine learning* (pp. 21486–21501). (vol. 235).
- Jin, Y., Song, X., Yang, Y., Hei, X., Feng, N., & Yang, X. (2025). An improved multi-channel and multi-scale domain adversarial neural network for fault diagnosis of the rolling bearing. *Control Engineering Practice*, 154, 106120. <https://doi.org/10.1016/j.conengprac.2024.106120>
- Ke, Z., Di, C., & Bao, X. (2022). Adaptive suppression of mode mixing in CEEMD based on genetic algorithm for motor bearing fault diagnosis. *IEEE Transactions on Magnetics*, 58(2), 1–6. <https://doi.org/10.1109/TMAG.2021.3082138>
- Li, Z., & Hoiem, D. (2017). Learning without forgetting. *arXiv:1606.09282*.
- Liang, Y.-L., Chien, W., & Yuan, S.-F. (2023). Overview of bearing fault diagnosis based on vibration signal. In *2023 IEEE 3rd international conference on electronic communications, internet of things and big data (ICEIB)* (pp. 547–549). <https://doi.org/10.1109/ICEIB57887.2023.10170527>
- Liu, Y., Chen, B., Wang, D., Kong, L., Shi, J., & Shen, C. (2023). A lifelong learning method based on generative feature replay for bearing diagnosis with incremental fault types. *IEEE Transactions on Instrumentation and Measurement*, 72, 1–11. <https://doi.org/10.1109/TIM.2023.3265739>
- Liu, Z., He, X., Huang, B., & Zhou, D. (2025). Incremental learning-enabled fault diagnosis of dynamic systems: A comprehensive review. *IEEE Transactions on Cybernetics*, 55(12), 5633–5649. <https://doi.org/10.1109/TCYB.2025.3586643>
- Loke, B. S. Y., Quadri, F., Vivanco, G., Casagrande, M., & Fenollosa, S. (2017). Overcoming catastrophic forgetting in neural networks. <https://arxiv.org/abs/2507.10485>.
- Lopez-Paz, D., & Ranzato, M. (2017). Gradient episodic memory for continual learning. In *Neural information processing systems*. <https://api.semanticscholar.org/CorpusID:37308416>.
- Shahzad, A., Liu, F., Jiang, M., Zhang, F., & Gan, Z. (2025). Multiscale feature fusion by entropy-augmented KCFP for rolling bearing fault diagnosis. *IEEE Sensors Journal*, 25(14), 26421–26431. <https://doi.org/10.1109/JSEN.2025.3573497>
- Song, B., Liu, Y., Fang, J., Liu, W., Zhong, M., & Liu, X. (2024). An optimized CNN-biLSTM network for bearing fault diagnosis under multiple working conditions with limited training samples. *Neurocomputing*, 574, 127284. <https://doi.org/10.1016/j.neucom.2024.127284>
- Song, Q., Jiang, X., Wang, S., Guo, J., Huang, W., & Zhu, Z. (2022). Self-adaptive multi-variate variational mode decomposition and its application for bearing fault diagnosis. *IEEE Transactions on Instrumentation and Measurement*, 71, 1–13. <https://doi.org/10.1109/TIM.2021.3139660>
- Sun, K., Xiao, B., Liu, D., & Wang, J. (2019). Deep high-resolution representation learning for human pose estimation. In *2019 IEEE/CVF Conference on computer vision and pattern recognition (CVPR)* (pp. 5686–5696). <https://doi.org/10.1109/CVPR.2019.00584>
- Wang, J., Sun, K., Cheng, T., Jiang, B., Deng, C., Zhao, Y., Liu, D., Mu, Y., Tan, M., Wang, X., Liu, W., & Xiao, B. (2021). Deep high-resolution representation learning for visual recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(10), 3349–3364. <https://doi.org/10.1109/TPAMI.2020.2983686>
- Wang, P., Xiong, H., & He, H. (2023). Bearing fault diagnosis under various conditions using an incremental learning-based multi-task shared classifier. *Knowledge-Based Systems*, 266, 110395. <https://doi.org/10.1016/j.knosys.2023.110395>
- Wang, Q., He, Y., Dong, S., Gao, X., Wang, S., & Gong, Y. (2025a). Non-exemplar domain incremental learning via cross-domain concept integration. In *Proc. IEEE/CVF conf. comput. vis* (pp. 144–162).
- Wang, Q., Song, X., He, Y., Han, J., Ding, C., Gao, X., & Gong, Y. (2025b). Boosting domain incremental learning: Selecting the optimal parameters is all you need. In *2025 IEEE/CVF Conference on computer vision and pattern recognition (CVPR)* (pp. 4839–4849). <https://doi.org/10.1109/CVPR52734.2025.00456>
- Wang, Z., Wu, Y., Wang, R., Lin, H., Wang, Q., Zhao, Q., & Meng, D. (2025c). Singular value fine-tuning for few-shot class-incremental learning. *ArXiv, abs/2503.10214*. <https://api.semanticscholar.org/CorpusID:276961064>.
- Xu, J., Mo, S., Jiang, Z., Chen, Z., Gui, W., & Wang, H. (2024a). A novel positive-negative graph convolutional network-based fault diagnosis method with application to complex systems. *Neurocomputing*, 600, 128145. <https://doi.org/10.1016/j.neucom.2024.128145>
- Xu, X., Bao, S., Liang, P., Qiao, Z., He, C., & Shi, P. (2024b). A broad learning model guided by global and local receptive causal features for online incremental machinery fault diagnosis. *Expert Systems with Applications*, 246, 123124. <https://doi.org/10.1016/j.eswa.2023.123124>
- Xu, Y., Deng, Y., Zhao, J., Tian, W., & Ma, C. (2020). A novel rolling bearing fault diagnosis method based on empirical wavelet transform and spectral trend. *IEEE Transactions on Instrumentation and Measurement*, 69(6), 2891–2904.
- Yu, Z., Zhang, C., & Deng, C. (2023). An improved GNN using dynamic graph embedding mechanism: A novel end-to-end framework for rolling bearing fault diagnosis under variable working conditions. *Mechanical Systems and Signal Processing*, 200, 110534. <https://doi.org/10.1016/j.ymssp.2023.110534>
- Zhang, Y., Shen, C., Shi, J., Li, C., Lin, X., Zhu, Z., & Wang, D. (2024). Deep adaptive sparse residual networks: A lifelong learning framework for rotating machinery fault diagnosis with domain increments. *Knowledge-Based Systems*, 293, 111679. <https://doi.org/10.1016/j.knosys.2024.111679>
- Zhou, Y., Dong, Y., Zhou, H., & Tang, G. (2021). Deep dynamic adaptive transfer network for rolling bearing fault diagnosis with considering cross-machine instance. *IEEE Transactions on Instrumentation and Measurement*, 70, 1–11. <https://doi.org/10.1109/TIM.2021.3112800>
- Zhu, W., Miao, Q., Cao, Y., Huang, P., & Fan, H. (2025). An early fault online detection model of rolling bearing based on deep attention convolutional autoencoder and multi-decision fusion under variable operation conditions. *Measurement*, 253, 117752. <https://doi.org/10.1016/j.measurement.2025.117752>